

7/24/26, 9:11 PM

Student Submissions — Analytical

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Student 17 **Kondam Bhaskar Tharunika** 192525276RC

5 / 5 submitted

Q1. Harshad number

```
ANSWER
n = int(input("Enter a number: "))
temp = n
s = 0

while temp > 0:
    s += temp % 10
    temp //= 10

if n % s == 0:
    print(n, "is a Harshad Number")
else:
    print(n, "is not a Harshad Number")

print("Harshad numbers from 1 to 500:")
for i in range(1, 501):
    temp = i
    s = 0
    while temp > 0:
        s += temp % 10
        temp //= 10
    if i % s == 0:
        print(i, end=" ")
```

Q2. GCD

CODE

```
a = int(input("Enter first: "))
b = int(input("Enter second: "))

while b != 0:
    temp = a
    a = b
    b = temp % b

print("GCD is:", a)
```

INPUT

12
18

OUTPUT

Enter first: Enter second: GCD is: 6

Q3. sum to a target value

```
ANSWER
def find_pairs(lst, target):
    seen = set()
    pairs = []

    for num in lst:
        complement = target - num
        if complement in seen:
            pairs.append((complement, num))
            seen.add(num)

    return pairs

data = [int(x) for x in input("Enter numbers: ").split()]
t = int(input("Target: "))

result = find_pairs(data, t)
print("Pairs:", result)
```

7/24/26, 9:11 PM

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Q4. Bitwise Operators

```
ANSWER
a = 0b1101
b = 0o25
c = 0x1A
```

```
data = [int(x) for x in input("Enter numbers: ").split()]
t = int(input("Target: "))

result = find_pairs(data, t)
print("Pairs:", result)
```

7/24/26, 9:11 PM

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Q4. Bitwise Operators

ANSWER

```
a = 0b1101
b = 0o25
c = 0x1A

result = (a & b) | c a = 0b1101
b = 0o25
c = 0x1A

result = (a & b) | c a = 13
b = 21
c = 26
result = 31
```

Q5. binary search

CODE

```
def binary_search(arr, key):
    low = 0
    high = len(arr) - 1

    while low <= high:
        mid = (low + high) // 2
        if arr[mid] == key:
            return mid
        elif arr[mid] < key:
            low = mid + 1
        else:
            high = mid - 1

    return -1

arr = list(map(int, input("Enter sorted numbers: ").split()))
key = int(input("Enter key: "))

result = binary_search(arr, key)

if result != -1:
    print("Element found at index", result)
else:
    print("Element not found")
```

INPUT

```
1 2 3 4 5 6 7 8 9
5
```

OUTPUT Enter sorted numbers: Enter key: Element found at index 4